

Riley Leff

PhD Candidate, Biological Sciences

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Education & Research Roles

PhD, Biological Sciences

2021 – Present

George Washington University, Washington, DC

- Research Interests: Plant Ecophysiology, Climate Change, Hydraulics, Compression, Intelligence, Open Source
- Research Advisor: Dr. Keryn Gedan

Research Associate

2019 - 2021

Pacific Northwest National Laboratory, Richland, WA

- Studied Tree Mortality In Response To Drought And Sea Level Rise
- Research Advisor: Dr. Nate McDowell

BSc, Biological Sciences

2015 – 2019

University of Chicago, Chicago, IL

- Honors B.S. Thesis: Microbial Diversity and Function in the Phyllosphere
- Research Advisor: Dr. Joy Bergelson

Software

Rust

- myco: a declarative, acausal programming language for scientific models. [\[Link\]](#)
- ozzydb: a purely functional approach to version control for scientific data. [\[Link\]](#)
- sdi12: no_std implementation of the SDI-12 protocol for embedded environmental sensors. The core runs anywhere with a GPIO, with adapters for RP2040, STM32, and ESP32. [\[Link\]](#)
- riley web services: self-hosted suite powering rileyleff.com, including `riley_auth` (accounts and sign-in), `riley_cms` (Git-as-database content management), `riley_leaderboards` (versioned rankings with live updates), and `riley_comments`. [\[Link\]](#)
- Floco: a library to enforce user-defined constraints on arbitrarily-sized floats directly within Rust's type system; an early inspiration for myco. [\[Link\]](#)
- Dirgrab: CLI utility for injecting entire local projects into LLM context windows. [\[Link\]](#)
- licor: Type-safe parser for LI-COR 6800 data with Python and R bindings. [\[Link\]](#)

Python

- quadrat: causal ecosystem simulator for benchmarking ecological reasoning in LLM agents, with Docker-sandboxed agent evaluation. Code available on request.
- rileyviewer: Browser-based plot viewer for matplotlib/seaborn, inspired by R's `httpgd`. [\[Link\]](#)

R

- ezSperry: Wrapper package to call a stomatal optimization model from R. [\[Link\]](#)

JavaScript / TypeScript

- Forest Royale: an educational 3D multiplayer game to communicate plant physiology concepts to students. [\[Link\]](#)
- annopurr: WebGPU image annotation tool for scientific measurements with TIFF support. [\[Link\]](#)
- rileaf2: PWA for measuring leaf area from images, designed for field use with the LI-COR 6800. [\[Link\]](#)
- RiLeaf.ijm: ImageJ macro automating leaf area quantification. [\[Link\]](#)

Hardware

- Sap flux logic board: open hardware for experimental heat-pulse sap flux sensors, built for a new method that solves for thermal parameters and probe positions conventional workflows assume; prototypes fabricated 2026. [Link]
- SDI-12 converter board: interface hardware (designed in atopile) connecting SDI-12 environmental sensors to any microcontroller with a GPIO. [Link]
- LI-6800 leaf clip: calibrated, 3D-printed leaf clip with a fiducial locator constellation and a companion offline phone app for in-chamber leaf area measurement.

Awards

Hidden Science Competition Winner	<i>Astera Institute</i>	2026
Dean's Dissertation Completion Fellowship	<i>GWU</i>	2026
Organismal Systems Research Exchange Grant	<i>University of Oxford</i>	2025
NSF Graduate Research Fellowship	<i>National Science Foundation</i>	April 2023
Harlan Graduate Research Fellowship	<i>GWU Biology</i>	April 2021, 2022, 2024, 2026
Irish Endowment Fund	<i>GWU Biology</i>	April 2023, 2024
Laboratory Mission Award Finalist	<i>Pacific Northwest National Laboratory</i>	September 2020
Research Honors	<i>UChicago Biological Sciences</i>	June 2019
Academic Honors	<i>The University of Chicago</i>	June 2019
Excellence in Teaching Award	<i>UChicago Biological Sciences</i>	June 2018
Ecology & Evolution Research Fellowship	<i>UChicago Biological Sciences</i>	May 2018
Dean's List	<i>The University of Chicago</i>	2016, 2017, 2018
The University Scholarship	<i>The University of Chicago</i>	March 2015

Publications & Datasets

Citations: 351 h-index: 9 [Link to Google Scholar](#)

Journal Articles

1. Biogeochemistry of upland to wetland soils, sediments, and surface waters across Mid-Atlantic and Great Lakes coastal interfaces

Allison N. Myers-Pigg, Stephanie C. Pennington, Khadijah K. Homolka, Allison M. Lewis, Opal Otenburg, Kaizad F. Patel, Peter Regier, Madison Bowe, Maxim I. Boyanov, Nathan A. Conroy, Donnie J. Day, Cooper G. Norris, Edward J. O'Loughlin, Jesse Alan Roebuck, Lucie Stetten, Vanessa L. Bailey, Kenneth M. Kemner, Nicholas D. Ward, EXCHANGE Consortium, Silver Alford, Michael P. Back, Andy Baldwin, Jade Bolinger, Jacob A. Cianci-Gaskill, Matthew J. Cooper, Alex Demeo, Kyle Derby, Derek Detweiler, Suzanne Devres-Zimmerman, Erin Eberhard, Keryn Gedan, LeeAnn Haaf, Erin Johnson, Aliya Khan, Matthew L. Kirwan, Payton Kittaka, Erika Koontz, Adam Langley, Riley Leff, Scott Lerberg, Sairah Y. Malkin, Amy M. Marcarelli, Steven E. McMurray, Tyler Messerschmidt, Taylor C. Michael, Holly A. Michael, Elizabeth C. Minor, Brian Moye, Thomas J. Mozdzer, Scott Neubauer, Andrea Pain, Michael Philben, Evan Phillips, Dannielle Pratt, Lauren Sage, Daniel Sandborn, Stacy Smith, Alexander Smith, Samina Soin-Voshell, Bongkeun Song, Amanda Sprague-Getsy, Kari St. Laurent, Lorie Staver, Alice Stearns, Rebecca Swerida, Ethan J. Theuerkauf, Katherine Tully, Rodrigo Vargas, Elizabeth Watson, and Coreen Weilminster
Scientific Data, 2023 [DOI] [Link]

2. Coastal inundation regime moderates the short-term effects of sediment and soil additions on seawater oxygen and greenhouse gas dynamics: a microcosm experiment

Peter Regier, Nicholas D. Ward, Alex Izquierdo, Andrew H. Baldwin, Donnie Day, Julia McElhinny, Kaizad Patel, Rodrigo Vargas, Jianqiu Zheng, Exchange Consortium, and Allison Myers-Pigg
Frontiers in Marine Science, 2023 [DOI] [Link]

3. Draft Genome Sequences of Three Bacterial Species from Aquatic Habitats near Washington, DC

Riley T. Leff, Jonathan MacDougall, Christina Pavludi, Lausanne Oliver, Kaitlynn Slattery, Guinevere Lissner, and Jimmy H. Saw

Microbiology Resource Announcements, 2023 [\[DOI\]](#) [\[Link\]](#)

4. Severe declines in hydraulic capacity and associated carbon starvation drive mortality in seawater exposed Sitka-spruce (*Picea sitchensis*) trees

Wenzhi Wang, Peipei Zhang, Hongxia Zhang, Charlotte Grossiord, Stephanie C Pennington, Matthew J Norwood, Weibin Li, Alexandria L Pivovarov, Laura Fernández-de-Uña, Riley Leff, Steven B Yabusaki, Scott Waichler, Vanessa L Bailey, Nicholas D Ward, and Nate G McDowell

Environmental Research Communications, 2022 [\[DOI\]](#) [\[Link\]](#)

5. Plant genetic effects on microbial hubs impact host fitness in repeated field trials

Benjamin Brachi, Daniele Filiault, Hannah Whitehurst, Paul Darne, Pierre Le Gars, Marine Le Mentec, Timothy C. Morton, Envel Kerdaffrec, Fernando Rabanal, Alison Anastasio, Mathew S. Box, Susan Duncan, Feng Huang, Riley Leff, Polina Novikova, Matthew Perisin, Takashi Tsuchimatsu, Roderick Woolley, Caroline Dean, Magnus Nordborg, Svante Holm, and Joy Bergelson

Proceedings of the National Academy of Sciences, 2022 [\[DOI\]](#) [\[Link\]](#)

6. The influence of increasing atmospheric CO₂, temperature, and vapor pressure deficit on seawater-induced tree mortality

Weibin Li, Nate G. McDowell, Hongxia Zhang, Wenzhi Wang, D. Scott Mackay, Riley Leff, Peipei Zhang, Nicholas D. Ward, Matt Norwood, Steve Yabusaki, Allison N. Myers-Pigg, Stephanie C. Pennington, Alexandria L. Pivovarov, Scott Waichler, Chonggang Xu, Ben Bond-Lamberty, and Vanessa L. Bailey

New Phytologist, 2022 [\[DOI\]](#) [\[Link\]](#)

7. Declining carbohydrate content of Sitka-spruce trees dying from seawater exposure

Peipei Zhang, Nate G McDowell, Xuhui Zhou, Wenzhi Wang, Riley T Leff, Alexandria L Pivovarov, Hongxia Zhang, Pak S Chow, Nicholas D Ward, Julia Indivero, Steven B Yabusaki, Scott Waichler, and Vanessa L Bailey

Plant Physiology, 2021 [\[DOI\]](#) [\[Link\]](#)

8. Hydraulic architecture explains species moisture dependency but not mortality rates across a tropical rainfall gradient

Alexandria L. Pivovarov, Brett T. Wolfe, Nate McDowell, Bradley Christoffersen, Stuart Davies, L. Turin Dickman, Charlotte Grossiord, Riley T. Leff, Alistair Rogers, Shawn P. Serbin, S. Joseph Wright, Jin Wu, Chonggang Xu, and Jeffrey Q. Chambers

Biotropica, 2021 [\[DOI\]](#) [\[Link\]](#)

9. Seawater exposure causes hydraulic damage in dying Sitka-spruce trees

Hongxia Zhang, Xinrong Li, Wenzhi Wang, Alexandria L. Pivovarov, Weibin Li, Peipei Zhang, Nicholas D. Ward, Allison Myers-Pigg, Henry D. Adams, Riley Leff, Anzhi Wang, Fenghui Yuan, Jiabing Wu, Steve Yabusaki, Scott Waichler, Vanessa L. Bailey, Dexin Guan, and Nate G. McDowell

Plant Physiology, 2021 [\[DOI\]](#) [\[Link\]](#)

10. Changes in carbon and nitrogen metabolism during seawater-induced mortality of *Picea sitchensis* trees

Weibin Li, Hongxia Zhang, Wenzhi Wang, Peipei Zhang, Nicholas D Ward, Matt Norwood, Allison Myers-Pigg, Chuanyan Zhao, Riley Leff, Steve Yabusaki, Scott Waichler, Vanessa L Bailey, and Nate G McDowell

Tree Physiology, 2021 [\[DOI\]](#) [\[Link\]](#)

11. Stability of tropical forest tree carbon-water relations in a rainfall exclusion treatment through shifts in effective water uptake depth

Alexandria L. Pivovarov, Nate G. McDowell, Tayana Barrozo Rodrigues, Tim Brodribb, Lucas A. Cernusak, Brendan Choat, Charlotte Grossiord, Yoko Ishida, Kolby J. Jardine, Susan Laurance, Riley Leff, Weibin Li, Michael Liddell, D. Scott Mackay, Heather Pacheco, Jennifer Peters, Israel J. Sampaio Filho, Daisy C. Souza, Wenzhi Wang, Peipei Zhang, and Jeff Chambers

Global Change Biology, 2021 [\[DOI\]](#) [\[Link\]](#)

12. Tree growth, transpiration, and water-use efficiency between shoreline and upland red maple (*Acer rubrum*) trees in a coastal forest

Wenzhi Wang, Nate G. McDowell, Stephanie Pennington, Charlotte Grossiord, Riley T. Leff, Aditi Sengupta, Nicholas D. Ward, Ugur Uzun Sezen, Roy Rich, J. Patrick Megonigal, James C. Stegen, Ben Bond-Lamberty, and Vanessa Bailey
Agricultural and Forest Meteorology, 2020 [\[DOI\]](#) [\[Link\]](#)

Published Datasets

EXCHANGE Campaign 1: A Community-Driven Baseline Characterization of Soils, Sediments, and Water Across Coastal Gradients

Stephanie C. Pennington, Silver Alford, Michael P. Back, Vanessa Bailey, Andy Baldwin, Jade Bolinger, Madison Bowe, Maxim I. Boyanov, Jacob A. Cianci-Gaskill, Nathan A. Conroy, Matthew J. Cooper, Donnie Day, Alex Demeo, Kyle Derby, Derek Detweiler, Suzanne Devres-Zimmerman, Erin Eberhard, Keryn Gedan, LeeAnn Haaf, Khadijah K. Homolka, Erin Johnson, Kenneth M. Kemner, Aliya Khan, Matthew Kirwan, Payton Kittaka, Erika Koontz, Adam Langley, Riley Leff, Scott Lerberg, Allison M. Lewis, Sairah Malkin, Amy M. Marcarelli, Steven E. McMurray, Tyler Messerschmidt, Taylor C. Michael, Holly A. Michael, Elizabeth C. Minor, Brian Moye, Thomas J. Mozdzer, Scott Neubauer, Cooper G. Norris, Edward J. O'Loughlin, Opal Otenburg, Andrea Pain, Kaizad F. Patel, Michael Philben, Evan Phillips, Dannielle Pratt, Peter Regier, Jesse Alan Roebuck Jr., Lauren Sage, Daniel Sandborn, Stacy Smith, Alex Smith, Samina Soin-Voshell, Bongkeun Song, Amanda Sprague-Getsy, Kari St. Laurent, Lorie Staver, Alice Stearns, Lucie Stetten, Rebecca Swerida, Ethan J. Theuerkauf, Katherine Tully, Rodrigo Vargas, Nicholas D. Ward, Elizabeth Watson, Coreen Weilminster, and Allison N. Myers-Pigg

Environmental System Science Data Infrastructure for a Virtual Ecosystem, 2023 [\[DOI\]](#) [\[Link\]](#)

Leaf gas exchange (BNL-PNNL collaboration), San Lorenzo, Panama, 2020

Julien Lamour, Kenneth Davidson, Kim Ely, Alexandria Pivovarov, Nate McDowell, Riley Leff, Alistair Rogers, and Shawn Serbin

Next-Generation Ecosystem Experiments Tropics; Brookhaven National Laboratory; Pacific Northwest National Laboratory, 2022 [\[DOI\]](#) [\[Link\]](#)

Functional trait data for 11 canopy tree species in the control and drought treatments at AU-DRO (2018-2019)

Alexandria Pivovarov, Nate McDowell, Tayana Barrozo, Timothy Brodribb, Lucas Cernusak, Charlotte Grossiord, Yoko Ishida, Kolby Jardine, Susan Laurance, Riley Leff, Weibin Li, Heather Pacheco, Israel Sampaio, Daisy Souza, Wenzhi Wang, and Peipei Zhang

Next-Generation Ecosystem Experiments Tropics; Pacific Northwest National Laboratory, 2021 [\[DOI\]](#) [\[Link\]](#)

Presentations

Talks

We Fixed Sap Flux

UVA Coastal Research Center Seminar, 2026

What Do The Trees Know?

George Mason University Computational Social Sciences Seminar, 2026

What Scientists Need To Know About AI

GWU Biology Department Seminar, 2025

Foliar Water Uptake Keeps Dying Trees Alive

Ecological Society of America Annual Meeting, 2025

Tree Mortality In Response To Seawater Inundation

George Washington University Seminar Series, 2023

Predicting Tree Function and Mortality in a Changing Tropical Environment

American Geophysical Union Fall Meeting, 2020

Microbial Diversity and Function in the Phyllosphere

University of Chicago Honors Research Symposium, 2019

Posters

Foliar Water Uptake Keeps Dying Trees Alive In A Coastal Forest

VCR LTER All-Hands Meeting, 2024

Reverse sap flow in coastal forest in response to saltwater flooding

New Phytologist Next-Gen Scientists, 2024

Hydraulic Stress Limits Pinus taeda Range In A Mid-Atlantic Coastal Forest

SALT Conference, 2024

Sap Flux: Updates After 3 Years Of Monitoring Across The Delmarva

Critical Zone Network All-Hands Meeting, 2024

Seawater Inundation Impacts On Pine Hydraulic Regime

Delaware Wetlands Conference, 2024

A Novel Quantitative Method To Improve Sap Flux Measurements

George Washington University Harlan Symposium, 2023

Wireless Sensor Networks For Automated Field Data Collection

VCR LTER All-Hands Meeting & Critical Zone Network Meeting, 2023

Seawater Intrusion Disrupts Sap Flux Dynamics In A Coastal Forest

George Washington University Harlan Symposium, 2022

Seawater Inundation Induces Reverse Sap Flux In Coastal Forest

Virginia Coastal Reserve LTER NSF Mid-Term Review, 2022

Predicting Tree Function and Mortality in a Changing Tropical Environment

American Geophysical Union Fall Meeting, 2020

Microbial Diversity and Function in the Phyllosphere

University of Chicago Honors Research Symposium, 2019

Teaching

Instructor: Setting Up Your Dev Machine For The Agent Era (Seminar)	2026
<i>UVA Coastal Research Center</i>	
Teaching Assistant: Conservation Biology	2026
<i>George Washington University</i>	
Instructor: Intro To Python (Summer Seminar Series)	2023
<i>George Washington University</i>	
Instructor: Intro To R (Summer Seminar Series)	2023
<i>George Washington University</i>	
Instructor: Intro To Ecology (Summer Seminar Series)	2022
<i>George Washington University</i>	
Teaching Assistant: Conservation Biology	2022
<i>George Washington University</i>	
Lab Instructor: Introductory Microbiology	2021
<i>George Washington University</i>	
Interim Instructor (7th Grade)	2020
<i>Twin Rivers Charter Academy, Yuba City, CA</i>	

Lab Instructor: Core Biology <i>University of Chicago</i>	2017 - 2019
Instructor / Mentor: Neighborhood Schools Program (5th Grade) <i>University of Chicago Outreach, Chicago, IL</i>	2016 - 2017
Corps Member / Tutor: Jumpstart (Pre-K Literacy) <i>University of Chicago Community Service Center, Chicago, IL</i>	2015 - 2016

Outreach & Engagement

Reverse sap flow in coastal forest in response to saltwater flooding Presentation at GWU Earth Day Symposium	2024
What Do The Trees Know? Activity at Virginia Coastal Reserve LTER 'Day In The Lab'	2024
Trees As Strategists Presentation at GWU Earth Day Symposium	2025
Trees As Strategists Activity at GWU Biology Graduate Student Council Science Fair	2025
Forest Royale Online Educational Game A 3D multiplayer plant physiology strategy game designed to teach students about complex tradeoffs in plant-environment interactions. <i>[Link]</i>	2025

Mentorship

Acadia Conboy Undergraduate, James Madison University - Project: Detailed photosynthetic characterization of <i>Phragmites australis</i> across a salinity gradient. - Outcome: Presented her work at two research symposia.	2026
Shay Cihlar Undergraduate, UC Davis - Project: Measurement and analysis of pressure-volume curves across salinity gradients - Outcome: Presented his work at two research symposia.	2025
Sehrin Hossain Undergraduate, George Washington University - Project: Automated image segmentation pipeline to analyze wood traits - Outcome: Won Luther Rice Fellowship, presented her work as a poster at a research symposium.	2025 – Present
Ellyn Kinkel Undergraduate, George Washington University - Project: Analyzing non-structural carbohydrate storage in salt-stressed coastal forests. - Outcome: Won Harlan Undergraduate Fellowship, presented her work as a poster at a research symposium.	2024 – Present
Kaeden Rippon Undergraduate, University of Virginia - Project: Detected foliar water uptake in pine saplings. - Outcome: Presented his work as a talk at a research symposium.	2024
Noa Shapiro-Tamir REU Student, Oberlin College	2023

- Project: Measured leaf decomposition rates across a salinity gradient.
- Outcome: Presented her work as a talk at a research symposium.

Maggie Connolly

2022

Undergraduate, George Washington University

- Project: Studying plant hydraulic responses to elevated salinity.
- Outcome: Presented her work as a talk and poster at a research symposium.

Aaron Mendez

2022

High School Student, GWU School Without Walls

- Project: Analyzing non-structural carbohydrate levels in *Phragmites australis* for AP Research project.
- Outcome: Presented his work, now a student at SUNY.

Aidan Brown

2022 – 2025

Master's Student (former undergraduate mentee), George Washington University

- Project: Training deep learning model to detect and analyze tree canopy structures from drone images.
- Outcome: Won GWU Harlan Fellowship, Critical Zone Network Fellowship (2x), Accepted to GWU 5-Year BS/MS Program In Biology.

James Abend

2019

High School Student, Tri Cities, WA

- Project: Processed samples for non-structural carbohydrate analysis.

Dakota DeRoos

2019

High School Student, Tri Cities, WA

- Project: Processed samples for non-structural carbohydrate analysis.